

# Ricardo Huaman

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## RESEARCH INTERESTS

Robot Learning, VLA, Computer Vision, Natural Language Processing, Deep Reinforcement Learning

## EDUCATION

**Universidad Privada Antenor Orrego (UPAO)**, Trujillo, Peru

*B.E. Electronic Engineering*

**July 2017 – July 2022**

- Class Rank: 1st
- Thesis supervisors: [Dr. Sixto Prado](#) & [Clayder Gonzalez, MSc](#)

## RESEARCH EXPERIENCE

**Cardiff University**, Cardiff, UK

*Research Assistant, Human-centred Computing Laboratory*

**January 2024 – April 2024**

- Advisors: [Dr. Carolina Fuentes](#) & [Dr. Fernando Alva](#)
- Through the [REPUcs 2024 Program](#), led the integration of foundational AI models with a humanoid robot, enhancing human-robot interaction in educational environments
- Implemented large language models (LLMs) into a quadruped robot simulation for autonomous indoor navigation, enabling more adaptive and intelligent behaviors in complex environments

**Universidad Privada Antenor Orrego**, Trujillo, Peru

*Research Assistant, Multidisciplinary Research Laboratory*

**May 2021 – December 2023**

- Advisors: [Dr. Sixto Prado](#) & [Clayder Gonzalez, MSc](#)
- I worked on the development of an agricultural mobile robot's autonomous navigation system, incorporating SLAM, path planning, and motion control algorithms. The components were integrated using the ROS middleware. This project aimed to enhance automation and productivity in the agricultural sector. Supported by government grant 171-2020-FONDECYT
- I worked on creating computer vision pipelines using deep learning to address specific agricultural challenges. Some examples include artichoke seedling classification, pre-harvest blueberry ripeness categorization, and the detection of missed seeding in plug trays. Supported by government grant 79170-2022-PROCIENCIA

## INDUSTRY EXPERIENCE

**Banco de Crédito BCP**, Lima, Peru

*Machine Learning Engineer*

**October 2025 – Present**

- Co-built and optimized an internal framework for the continuous monitoring of 40+ production models spanning consumer, SME, and wholesale banking segments, safeguarding the operational reliability of business-critical models across the bank
- Migrated and managed 15+ analytical model pipelines into production for consumer banking under tight deadlines, powering bank-wide marketing campaigns that reach millions of clients
- Co-lead an internal Advanced Analytics COE knowledge-sharing program, coordinating monthly technical sessions that reach 100+ Machine Learning Engineers, Data Scientists, and Analytics Translators

**IP Noticias LATAM**, Lima, Peru

*Artificial Intelligence Engineer*

**May 2024 – September 2025**

- Built object detection and real-time facial recognition systems for streaming media using gRPC for high-efficiency inter-service communication, cutting manual processing time by 60% and identifying over 1,000 individuals across multimedia content
- Designed a scalable, microservices-based infrastructure to deploy text classification and named entity recognition (NER) models, optimizing inference speed across 50+ models by applying MLOps best practices
- Developed LLM-powered RAG agents (LangChain, Agno, vector/relational databases) and an embedding-based thematic discovery pipeline, delivering context-aware responses and automated topic identification across large volumes of text

- Curated and annotated computer-vision datasets of hundreds of images, building preprocessing pipelines with Python and OpenCV to support model training
- Implemented and trained Deep Learning models in PyTorch for image classification and detection tasks, adapting architectures to the requirements of each project
- Performed exploratory data analysis and data-quality checks to improve dataset reliability across research projects

SELECTED  
PUBLICATIONS

- [1] **R. J. H. Kemper**, P. A. Zamora Manrique, D. D. Herrera Rosas and S. R. Prado Gardini, “Performance Evaluation of Detection and Tracking Algorithms for Automated Grape Cluster Counting,” 2025 IEEE Colombian Caribbean Conference, 2025, pp. 1-6, doi: [10.1109/C366505.2025.11340275](https://doi.org/10.1109/C366505.2025.11340275)
- [2] **R. Huaman**, C. Gonzalez and S. Prado, “Comparative Analysis of LiDAR Inertial Odometry Algorithms in Blueberry Crops,” Engineering Proceedings, vol. 83, no. 1, 2025, doi: [10.3390/eng-proc2025083009](https://doi.org/10.3390/eng-proc2025083009)
- [3] **R. Huaman**, C. Gonzalez and S. Prado, “Performance Evaluation of the ROS Navigation Stack using LeGO-LOAM,” Proceedings of the 9th Brazilian Technology Symposium (BTSym’23), Lima, Peru, 2024, pp. 173-183, doi: [10.1007/978-3-031-66961-3\\_16](https://doi.org/10.1007/978-3-031-66961-3_16)
- [4] P. B. Cubas Muñoz, **R. J. Huaman Kemper** and S. R. Prado Gardini, “Artificial vision strategy for Ripeness assessment of Blueberries on Images taken during Pre-harvest stage in Agroindustrial Environments using Deep Learning Techniques,” 2023 IEEE XXX International Conference on Electronics, Electrical Engineering and Computing (INTERCON), Lima, Peru, 2023, pp. 1-6, doi: [10.1109/INTERCON59652.2023.10326058](https://doi.org/10.1109/INTERCON59652.2023.10326058)
- [5] P. B. Cubas Muñoz, **R. J. Huaman Kemper**, E. M. Fiestas Sorogastua and S. R. Prado Gardini, “Deep Learning-based Segmentation and Classification System for Artichoke Seedling Grading,” 2023 IEEE XXX International Conference on Electronics, Electrical Engineering and Computing (INTERCON), Lima, Peru, 2023, pp. 1-6, doi: [10.1109/INTERCON59652.2023.10326060](https://doi.org/10.1109/INTERCON59652.2023.10326060)
- [6] J. A. B. Quispe, **R. J. H. Kemper** and S. R. P. Gardini, “Linear Quadratic Regulator (LQR) Control for the Active Suspension System of a Four-Wheeled Agricultural Robot,” 2023 IEEE XXX International Conference on Electronics, Electrical Engineering and Computing (INTERCON), Lima, Peru, 2023, pp. 1-6, doi: [10.1109/INTERCON59652.2023.10326049](https://doi.org/10.1109/INTERCON59652.2023.10326049)
- [7] **R. J. H. Kemper**, C. Gonzalez and S. R. P. Gardini, “Autonomous Navigation of a Four-Wheeled Robot in a Simulated Blueberry Farm Environment,” 2022 IEEE ANDESCON, Barranquilla, Colombia, 2022, pp. 1-6, doi: [10.1109/ANDESCON56260.2022.9989865](https://doi.org/10.1109/ANDESCON56260.2022.9989865)

ACADEMIC  
TALKS AND  
POSTER  
SESSIONS

- P. Cubas\*, **R. Huaman\*** and S. Prado, “Deep Learning Approach for Accurate Pre-Harvest Blueberry Ripeness Classification”. Poster, *Workshop on Robotics in Agriculture at IROS*. October, 2023
- C. Gonzalez\*, **R. Huaman\*** and S. Prado, “Comparative Analysis of Lidar Odometry Algorithms in Blueberry Crops”. Poster, *Workshop on Robotics in Agriculture at IROS*. October, 2023
- R. Huaman**, C. Gonzalez, and S. Prado, “Integration of an Autonomous Navigation System for an Agricultural Mobile Robot in a Simulated Blueberry Cultivation Environment”. Talk, *XIII International Scientific Meeting of the North at UPAO*. December, 2022
- R. Huaman**, C. Gonzalez, and S. Prado, “Implementation of an Autonomous Navigation System for a Mobile Robot in a Simulated Environment of a Blueberry Agro-Industrial Farm”. Talk, *I International Congress of Research and Innovation Incubators at UPAO*. November, 2022
- R. Huaman**, “Path Planning for Autonomous Robots”. Talk, *Electronics Engineering Dept. at UPAO*. May, 2022

PROGRAMMING  
& SOFTWARE

Python, C++, PyTorch, TensorFlow, ROS 1 & 2, Git, Linux, MATLAB, AWS, Azure, Docker, Kubernetes, OpenCV, LangChain

## LANGUAGES

**Languages** Spanish (*native*), English (*TOEFL iBT 104/120*), Japanese (*JLPT N2*)